

RICHARD SIMMONS

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[Portfolio](#): richard-simmons-portfolio.vercel.app/ | [GitHub](#): github.com/RSimmons2021 | [LinkedIn](#): linkedin.com/in/richard-simmons-a3916958/

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), Java, C++, SQL

Frontend: React, Next.js, React Native, TailwindCSS, Three.js, Redux

Backend: Node.js, PostgreSQL, Redis, GraphQL, REST APIs, Flask, Microservices, Distributed Systems

Cloud/DevOps: AWS (S3, Lambda, EC2), Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins), Supabase

PROFESSIONAL EXPERIENCE

Founder & Lead Engineer

Zoan Collective | Columbus, OH | April 2025 – Present

- Shipped 2 production mobile apps (FocusFi, Lucid) to iOS App Store and Google Play with 500+ combined downloads; architected full-stack systems using React Native, Next.js, Supabase, and Stripe payment integration.
- Built LLM-driven learning and audio-generation pipelines processing 1,000+ user interactions/day through background workers, caching layers, structured outputs, and monitoring instrumentation.
- Built serverless backend architecture with Supabase Edge Functions handling AI music generation, payment processing (iOS StoreKit 2 & Google Play Billing), and rate limiting; reduced infrastructure costs by 80% vs traditional server deployment.
- Designed fault-tolerant authentication, subscription, and webhook processing systems supporting cross-platform mobile distribution.

Software Engineer

Toyota | Chicago, IL | June 2021 – March 2025

- Architected and deployed RESTful APIs and GraphQL services handling 1M+ daily requests across manufacturing analytics platform, supporting real-time data processing for 10,000+ daily active users across 7 North American facilities.
- Built 5+ production applications including inventory management systems and quality control dashboards using React/TypeScript frontend and Python/Flask/Node.js backend, reducing manual reporting time by 70% for mechanical engineering team.
- Reduced dashboard load time from 8s to 400ms (95% improvement) by redesigning query patterns across 15+ joined tables and implementing materialized views with Redis caching layer.
- Led UX audit across 3 enterprise modules serving 2,000+ users; redesigned navigation architecture reducing average task completion time by 35% and support tickets by 40%.
- Established Jest and Cypress testing suite achieving 85% code coverage; integrated automated testing into CI/CD pipeline, reducing production bugs by 60% and enabling daily deployments.

Data Science Intern

Knoesis Labs | Dayton, OH | June 2018 – August 2020

- Analyzed social media datasets exceeding 5M records using Python (Pandas, NumPy, Scikit-learn), identifying sentiment patterns with 87% classification accuracy for research publications.
- Engineered ETL pipelines using Azure Data Factory and SSIS processing 5GB+ daily data, reducing data preparation time by 60% for research team.

PROJECTS

FocusFi — AI-Powered Study & Productivity Platform

React Native, Next.js, TypeScript, Supabase, Stripe | [Website](#)

- Architected cross-platform study application with gamified learning system implementing spaced repetition (SM-2 algorithm), Pomodoro timer, and 7+ evidence-based learning techniques across iOS, Android, and web.
- Engineered real-time activity tracking with PostgreSQL logging 4 activity types and rendering interactive data visualizations including stacked bar charts, 90-day heatmaps, and streak analytics.

Lucid — AI Binaural Beats & Sleep Music Platform

React Native, Supabase Edge Functions, Suno AI | [Website](#)

- Developed personalized meditation-audio generation systems using prompt orchestration and background workers with secure payment webhook handling.

EDUCATION

Bachelor of Science in Computer Science

Wright State University | *Dayton, OH* | *May 2021*